Renal Reference — Urine Sediment Cheat Sheet

Urine Sediment Cheat Sheet

Urine sediment clue	Diagnosis suggested	Why
Bland sediment + hyaline casts	**Prerenal AKI**	No structural injury; low-flow/concentrated urine
Muddy brown granular casts	**ATN**	Tubular epithelial injury and debris
Renal tubular epithelial cells/casts	**ATN**	Tubular cells are sloughing
RBC casts	**Glomerulonephritis**	Bleeding from inflamed glomeruli
Dysmorphic RBCs	**GN / vasculitis**	RBCs are damaged while crossing the glomerulus
WBC casts	**AIN / pyelonephritis**	Interstitial or renal inflammation
Sterile pyuria	**AIN**	WBCs without bacterial infection
Heme-positive dipstick + few/no RBCs	**Rhabdo or hemolysis**	Myoglobin/hemoglobin detected as "heme"
Calcium oxalate crystals	**Ethylene glycol**	Oxalate crystal nephropathy
Uric acid crystals	**Tumor lysis syndrome**	Uric acid precipitation in tubules
Drug crystals	**Acyclovir / indinavir**	Intratubular crystal obstruction

1. Bland sediment + hyaline casts → Prerenal

WHAT YOU SEE

Bland sediment + hyaline casts → Prerenal AKI

WHAT IT ENCODES

A bland (acellular, cast-poor) sediment points to a functional, non-structural cause — prerenal azotemia from low effective volume or low cardiac output. Hyaline casts are pure Tamm-Horsfall (uromodulin) protein, are non-specific, and form in concentrated/low-flow urine, after exercise, or with diuretics. Prerenal labs: FENa <1%, urine osm >500, BUN:Cr >20:1; the AKI corrects with volume.

BOARD TRAP

WRONG: equating a bland sediment with ATN or assuming hyaline casts indicate disease. Bland sediment + hyaline casts = prerenal, NOT ATN (which shows muddy-brown granular casts). Discriminator: hyaline casts can be entirely normal in concentrated urine — they do not signify intrinsic injury.

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2. Muddy brown granular casts → ATN

WHAT YOU SEE

Muddy brown granular casts → ATN

WHAT IT ENCODES

Muddy-brown coarse granular ('dirty brown') casts are sloughed necrotic tubular epithelial cells and debris packed into a Tamm-Horsfall matrix — the hallmark of established ischemic or nephrotoxic ATN. A higher cast/RTE-cell score correlates with worse AKI and lower odds of recovery. Pairs with FENa >2% (off diuretics) and a non-oliguric or oliguric course.

BOARD TRAP

WRONG: calling muddy-brown casts evidence of glomerular disease. Muddy-brown granular casts are specific for tubular injury (ATN), not GN (which gives RBC casts). Discriminator: granular/pigmented casts + RTE cells + nephrotoxin or ischemic insult = ATN; treat supportively, not with immunosuppression.

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3. Renal tubular epithelial cells/casts → ATN

WHAT YOU SEE

Renal tubular epithelial cells / casts → ATN

WHAT IT ENCODES

Free renal tubular epithelial (RTE) cells and RTE-cell casts are directly sloughed dead tubular cells, confirming tubular necrosis. They co-occur with muddy-brown granular casts in ATN and, together, form a quantifiable urinary microscopy score that predicts AKI severity and dialysis need.

BOARD TRAP

WRONG: mistaking RTE cells for squamous epithelial cells (contamination) or dysmorphic RBCs. RTE cells are larger with a single large central nucleus and signify tubular necrosis; abundant squamous cells just mean a poorly collected specimen. Discriminator: RTE cells/casts localize injury to the tubule (ATN), not the glomerulus.

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4. RBC casts → GN

WHAT YOU SEE

RBC casts → Glomerulonephritis

WHAT IT ENCODES

Red-cell casts form when RBCs that bled through an inflamed glomerular basement membrane are molded into a tubular cast — the signature of glomerulonephritis/nephritic syndrome. Their presence mandates GN serologies (ANA/dsDNA, ANCA, anti-GBM, complement C3/C4, ASO, hepatitis/HIV, cryoglobulins) and usually kidney biopsy.

BOARD TRAP

WRONG: attributing hematuria with RBC casts to a stone, tumor, or bladder source. RBC casts are essentially pathognomonic for glomerular bleeding — a urologic (lower-tract) source does NOT produce casts. Discriminator: RBC casts + dysmorphic RBCs + proteinuria + HTN = GN; pursue serologies/biopsy, not just a CT urogram.

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5. Dysmorphic RBCs → GN / vasculitis

WHAT YOU SEE

Dysmorphic RBCs → GN / vasculitis

WHAT IT ENCODES

Dysmorphic (irregular, acanthocytic/'Mickey Mouse') RBCs are deformed as they squeeze through the damaged glomerular basement membrane, indicating a glomerular source of hematuria (GN/vasculitis). Acanthocytes (ring forms with membrane blebs) >5% are particularly specific for glomerular bleeding.

BOARD TRAP

WRONG: assuming any microscopic hematuria warrants a urologic malignancy workup first. Isomorphic (normal, uniform) RBCs point to a urologic/lower-tract source (stone, tumor, trauma), while dysmorphic RBCs point to the glomerulus. Discriminator: RBC morphology triages the hematuria — dysmorphic → nephrology/serologies; isomorphic → urology/imaging.

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6. WBC casts → AIN / pyelonephritis

WHAT YOU SEE

WBC casts → AIN / pyelonephritis

WHAT IT ENCODES

White-cell casts form when leukocytes from interstitial or renal parenchymal inflammation are packed into a cast, localizing the process to the upper tract/kidney. Seen in acute interstitial nephritis (drug-induced: PPIs, NSAIDs, beta-lactams, sulfa, checkpoint inhibitors) and in acute pyelonephritis.

BOARD TRAP

WRONG: interpreting pyuria with WBC casts as simple cystitis. WBC casts indicate UPPER-tract/renal inflammation, distinguishing pyelonephritis (or AIN) from lower-tract cystitis, which does not cast. Discriminator: WBC casts + fever/flank pain → pyelonephritis; WBC casts + new drug + sterile culture → AIN.

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7. Sterile pyuria → AIN

WHAT YOU SEE

Sterile pyuria → AIN

WHAT IT ENCODES

Sterile pyuria is WBCs in the urine with a NEGATIVE standard bacterial culture. In the AKI context it suggests acute interstitial nephritis (drug-induced), but the full differential is broad: renal TB, Chlamydia/gonorrhea and other urethritis, recently/partially treated UTI, fastidious organisms, nephrolithiasis, and contamination.

BOARD TRAP

WRONG: assuming sterile pyuria always equals AIN. Genitourinary TB and chlamydial urethritis are classic boards causes of sterile pyuria that get missed. Discriminator: in sterile pyuria, weigh AIN (new drug, eosinophils, WBC casts) against TB (risk factors, send AFB cultures) and STI (sexual history, NAAT) before settling on AIN.

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8. Heme+ dipstick + few/no RBCs → Rhabdo/hemolysis

WHAT YOU SEE

Heme-positive dipstick + few/no RBCs → Rhabdo or hemolysis

WHAT IT ENCODES

A heme-positive dipstick reflects peroxidase activity from heme moieties — RBCs, myoglobin, OR hemoglobin. When the dipstick is heme-positive but microscopy shows FEW/no RBCs, the cause is pigment: myoglobinuria (rhabdomyolysis — CK often >5,000) or hemoglobinuria (intravascular hemolysis — low haptoglobin, high LDH, plasma pink in hemolysis but not rhabdo).

BOARD TRAP

WRONG: treating heme-positive dipstick with no RBCs as glomerular hematuria. Heme+ without RBCs is pigment (myoglobin/hemoglobin), not RBC bleeding. Discriminator: check CK (rhabdo) and haptoglobin/LDH/plasma color (hemolysis); pigment nephropathy can also give a deceptively LOW FENa from early renal vasoconstriction.

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9. Calcium oxalate crystals → Ethylene glycol

WHAT YOU SEE

Calcium oxalate crystals → Ethylene glycol

WHAT IT ENCODES

Calcium oxalate crystalluria (envelope-shaped dihydrate or needle-shaped monohydrate) in the right context signals ethylene glycol (antifreeze) poisoning: high-anion-gap metabolic acidosis PLUS an elevated osmolar gap, with AKI from oxalate crystal nephropathy. Treat with fomepizole (alcohol dehydrogenase inhibitor) ± hemodialysis; sodium bicarbonate for acidosis.

BOARD TRAP

WRONG: chalking up oxalate crystals to dehydration or diet and missing the toxic alcohol. Calcium oxalate crystals + high anion-gap acidosis + osmolar gap = ethylene glycol toxicity (an emergency). Discriminator: methanol gives the same gaps but causes visual loss without prominent oxalate crystals — both need fomepizole ± dialysis.

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10. Uric acid crystals → Tumor lysis

WHAT YOU SEE

Uric acid crystals → Tumor lysis syndrome

WHAT IT ENCODES

Uric acid crystals (rhomboid/needle, often after cytotoxic therapy of bulky/high-grade lymphomas/leukemias) reflect tumor lysis syndrome: hyperuricemia, hyperkalemia, hyperphosphatemia, hypocalcemia, and AKI from intratubular urate (and calcium-phosphate) precipitation. Prevent/treat with aggressive IV fluids + rasburicase (high risk) or allopurinol (lower risk); avoid urine alkalinization when phosphate is high.

BOARD TRAP

WRONG: conflating uric acid AKI in TLS with chronic gouty/urate nephropathy. Acute uric acid crystal AKI = tumor lysis (prevent with hydration + rasburicase), a different entity from chronic gout nephropathy. Discriminator: a chemo/high-turnover-tumor setting with the TLS electrolyte panel points to TLS — start rasburicase, not just allopurinol, in high-risk patients.

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11. Drug crystals → Acyclovir / indinavir

WHAT YOU SEE

Drug crystals → Acyclovir / indinavir

WHAT IT ENCODES

Poorly soluble drugs precipitate as intratubular crystals causing crystalline AKI: acyclovir (needle-shaped, with rapid IV bolus + volume depletion), indinavir/atazanavir (plate/rosette crystals), sulfadiazine, methotrexate, and triamterene. Prevention/treatment is aggressive hydration (and slow infusion for IV acyclovir; urine alkalinization + leucovorin ± glucarpidase for methotrexate).

BOARD TRAP

WRONG: managing crystalline drug AKI like AIN with steroids. Crystal nephropathy (acyclovir/indinavir/sulfa/MTX) is an intratubular OBSTRUCTIVE injury treated with hydration and stopping/adjusting the drug — distinct from immune-mediated AIN. Discriminator: drug-specific crystals on microscopy + the offending agent + recovery with hydration confirm crystalline AKI rather than AIN.
